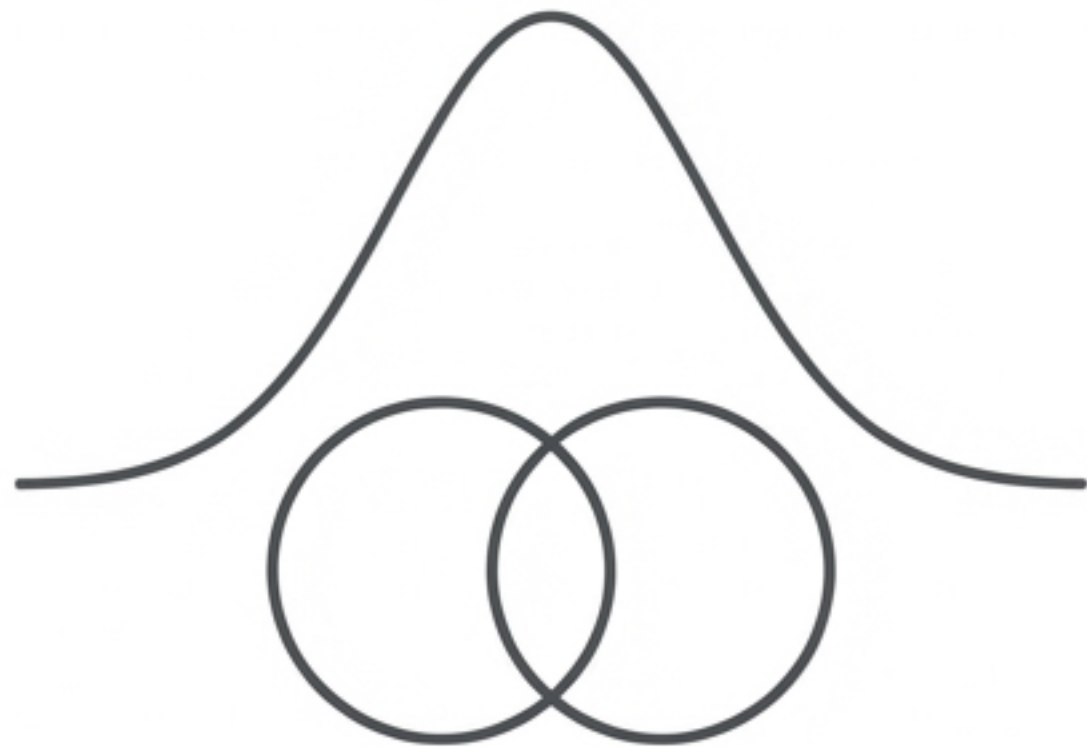




Probability & Inference

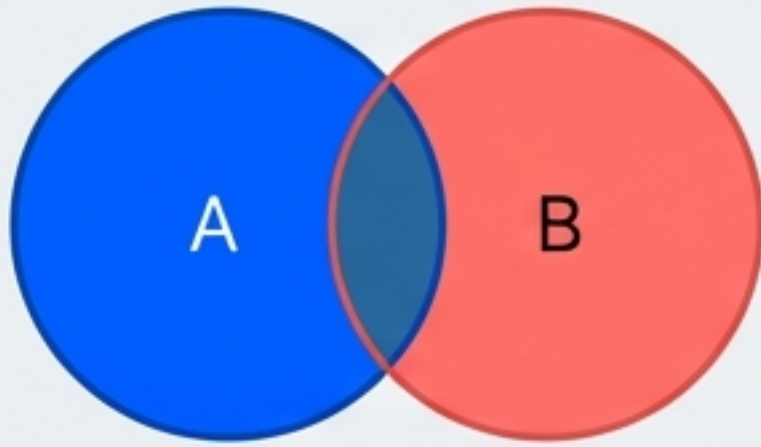
From Conditional Logic to Bayes' Theorem

A technical guide to understanding dependent events, filtering noise, and updating beliefs with evidence.

The Landscape: How Events Interact

To calculate probability accurately, we must first determine if events influence one another.

1. Independent Events

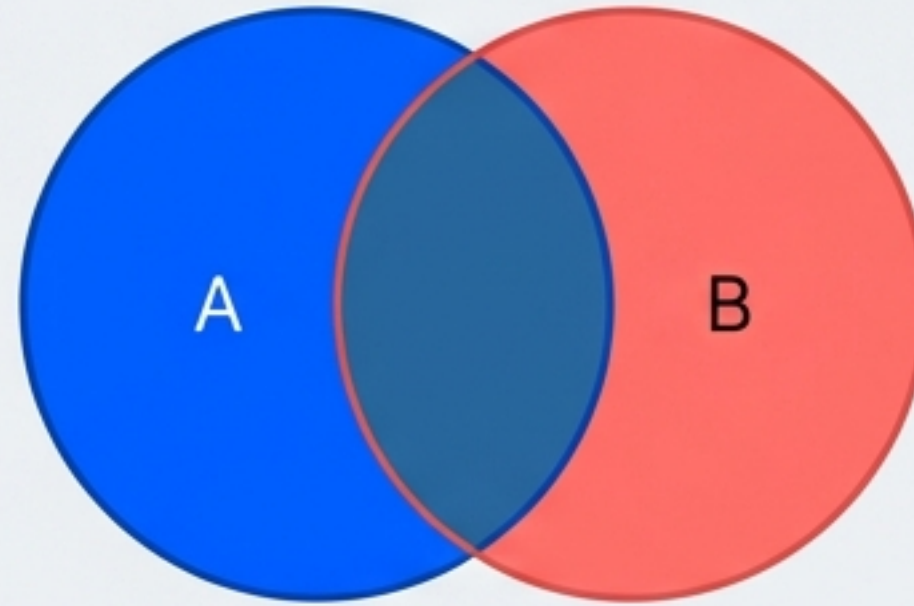


Occurrence of one does not affect the other.

$$P(A \cap B) = P(A) \times P(B)$$

Coin toss (Head) & Die roll (4).
Odds: 1/12.

2. Dependent Events

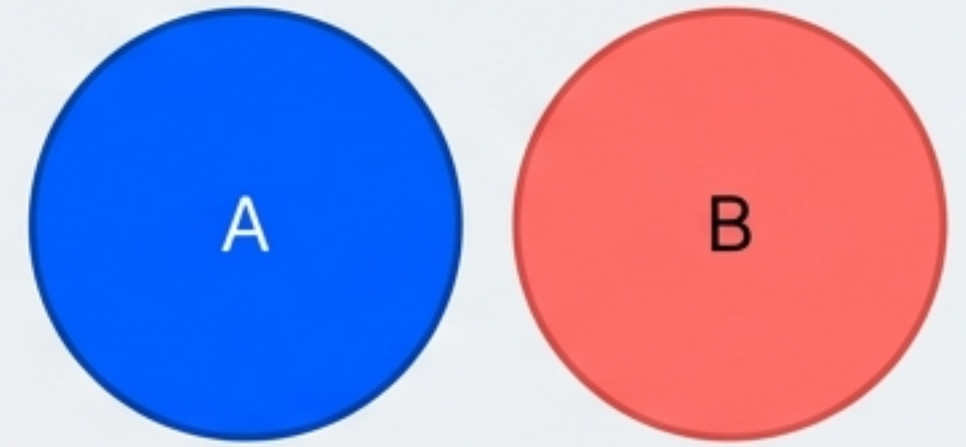


Occurrence of one changes the probability of the other.

$$P(A \cap B) = P(A) \times P(B|A)$$

Drawing Aces without replacement. Odds shift.

3. Mutually Exclusive



Cannot occur simultaneously.

$$P(A \cap B) = 0$$

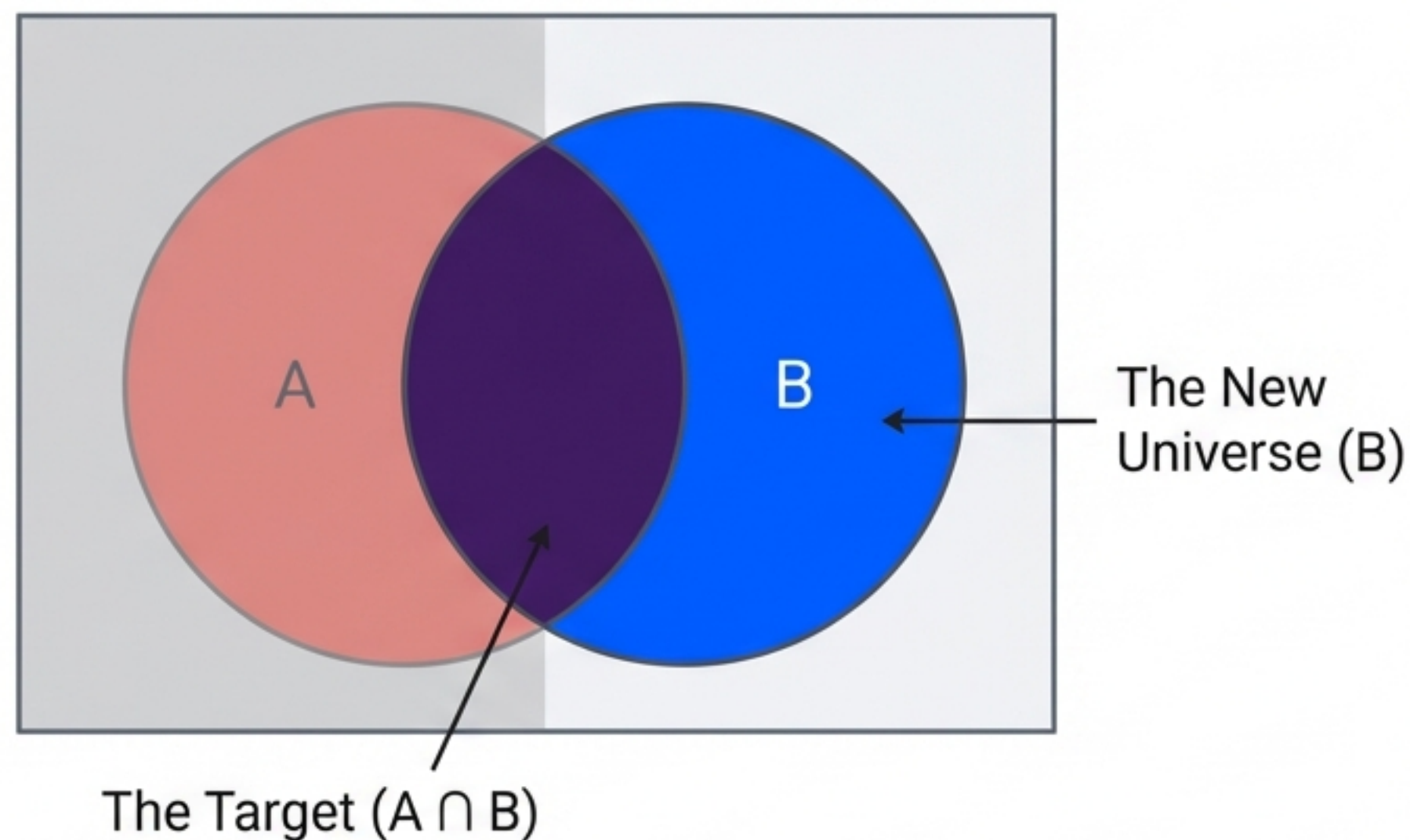
Rolling a 2 and a 5 at the same time.

Conditional Probability “Filters” the Sample Space

Definition: Conditional probability ($P(A|B)$) is the likelihood of event A occurring given that event B has already happened.

The Intuition: Think of event B as a filter. We stop looking at the “whole pie” (all possible outcomes) and restrict universe only to the “B slice.” We then ask: “Out of these specific B situations, what fraction also satisfies A?”

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



Visual Intuition: The Coffee & Tea Problem

The Scenario: 100 People total.

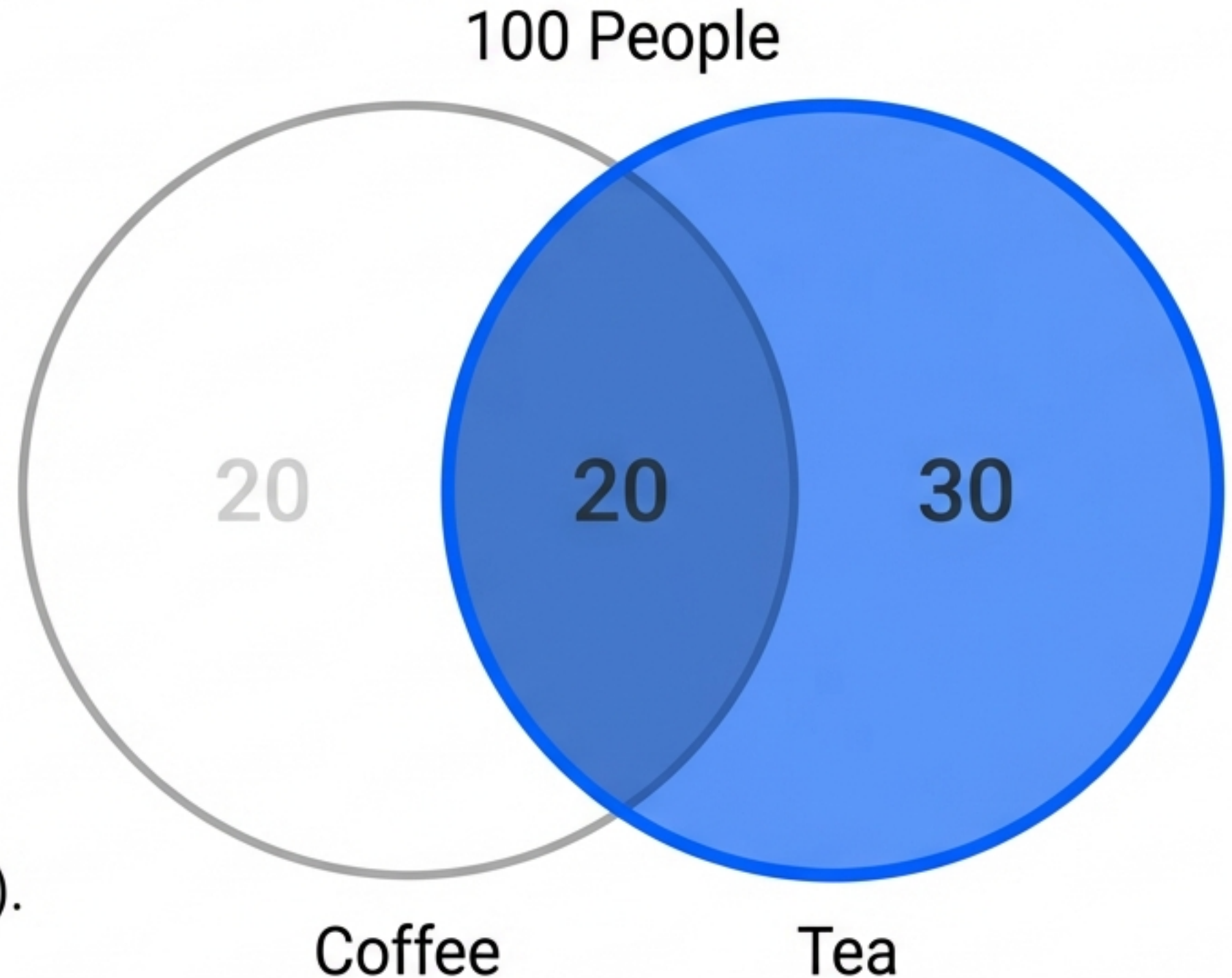
- 40 like Coffee (A)
- 50 like Tea (B)
- 20 like Both ($A \cap B$)

The Question:

If we select a person who likes Tea, what is the probability they also like Coffee?

The Calculation:

- Step 1: Filter to Tea drinkers (50 people).
Step 2: Identify overlap with Coffee (20 people).
Step 3: Result = $20 / 50 = 0.4$ (40%)



Applied Context: Segmenting Customer Data

	Premium	Basic	Total
USA	80	220	300
UK	120	580	700
Total	200	800	1000

P(Premium | USA)

Condition:

USA (Row Total = 300)

Intersection:

Premium & USA (80)

Calculation: $\frac{80}{300} \approx 26.7\%$

P(USA | Premium)

Condition:

Premium (Col Total = 200)

Intersection:

Premium & USA (80)

Calculation: $\frac{80}{200} = 40\%$

Key Insight: Changing the condition (denominator) dramatically flips the probability.

The Toolkit: Python Implementation

1. Build the DataFrame from raw data.
2. Calculate Row and Column totals (Marginals).
3. Compute Conditional Probability by dividing the specific intersection cell by the marginal total.

Intersection / Condition



```
import pandas as pd

# 1. Create the Contingency Table
data = {"Premium": [80, 120], "Basic": [220, 580]}
df = pd.DataFrame(data, index=["USA", "UK"])

# 2. Calculate Totals
df["Total"] = df.sum(axis=1)           # Row sums
df.loc["Total"] = df.sum()             # Column sums
N = df.loc["Total", "Total"]           # Grand total (1000)

# 3. Calculate Conditional Probability P(Premium | USA)
joint_premium_usa = df.loc["USA", "Premium"] / N  # P(A and B)
p_usa = df.loc["USA", "Total"] / N                 # P(B)

p_premium_given_usa = joint_premium_usa / p_usa
print(f"P(Premium | USA) = {p_premium_given_usa:.3f}")
```


Bayes' Theorem: Updating Beliefs with Evidence

Bayes allows us to reverse the condition: Given the evidence, how likely is the cause?

Likelihood

Probability of evidence B,
assuming A is true.

Prior

Probability of A before
new evidence.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Posterior

Updated probability of
A given B.

Evidence

The total probability of
B occurring.

Example 1: The Medical Test Paradox

Why a 99% accurate test result doesn't mean a 99% chance of disease.

Knowns

- **Prior (Disease Rate):**
 $P(A) = 0.01$
- **Likelihood (Test Accuracy):**
 $P(B|A) = 0.99$
- **False Positive Rate:**
 $P(B|\text{not } A) = 0.01$



The Calculation Stack

Step 1: The Evidence (Denominator)

Calculate $P(B)$ using Total Probability Theorem:

$$P(B) = (0.99 \times 0.01) + (0.01 \times 0.99) = 0.0198 \quad \text{(Evidence)}$$

Step 2: The Update

Apply Bayes' Theorem:

$$P(A|B) = \frac{0.99 \times 0.01}{0.0198} = 0.5 \quad \text{(Posterior)}$$



RESULT: 50%

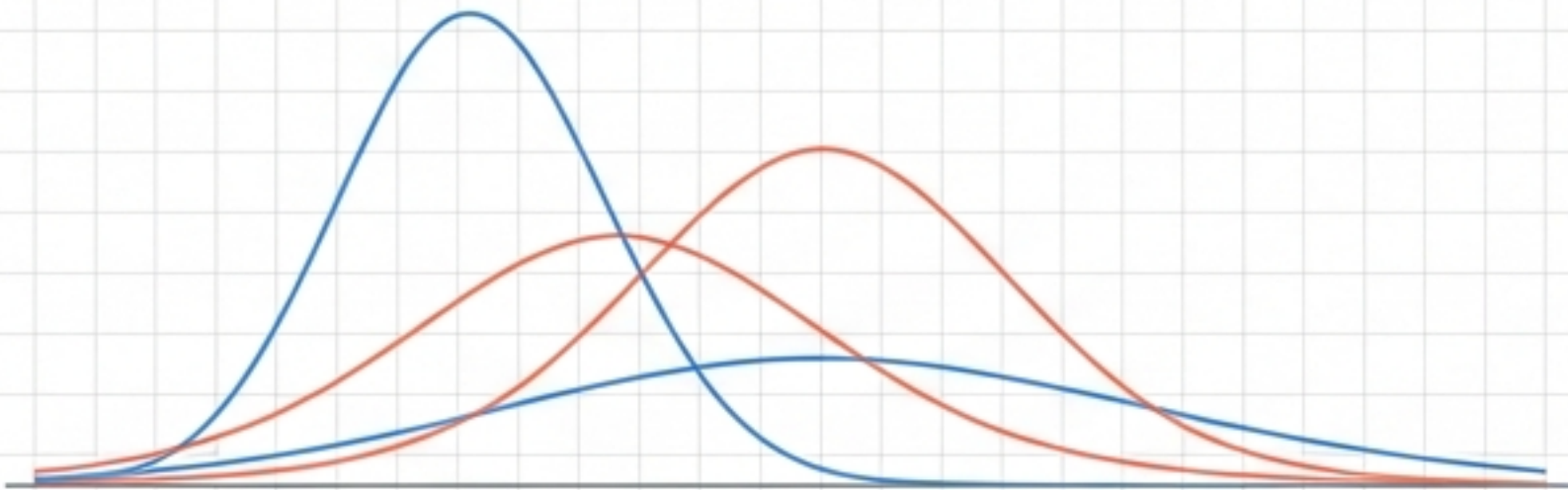
Since the disease is rare, the False Positives (from the healthy population) weigh heavily against the True Positives.

Example 2: Spam Detection Logic

Calculating the probability of an email being spam given it contains the word "Offer".

Knowns

- Prior (Spam Rate): $P(A) = 0.40$
- Likelihood ("Offer" in Spam): $P(B|A) = 0.20$
- Non-Spam Context ("Offer" in Ham): $P(B|\text{not } A) = 0.05$



The Calculation Stack

Step 1: The Evidence (Total emails with "Offer")

Calculate $P(B)$ using Total Probability Theorem:

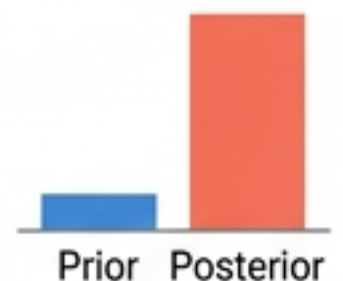
$$P(B) = (0.2 \times 0.4) + (0.05 \times 0.6)$$

$$P(B) = 0.08 + 0.03 = 0.11 \longrightarrow \text{(Evidence)}$$

Step 2: The Update

Apply Bayes' Theorem:

$$P(A|B) = \frac{0.08}{0.11} \approx 0.727 \longrightarrow \text{(Posterior)}$$



RESULT: 72.7%

If an email contains the word 'Offer', the probability it is spam jumps from 40% to 72.7%.

Coding the Belief Updater

A reusable Python function for Bayesian inference.

```
def bayes_theorem(p_a, p_b_given_a, p_b_given_not_a):  
    # 1. Calculate Complement of Prior  
    p_not_a = 1 - p_a  
  
    # 2. Calculate Marginal Probability (Evidence)  
    p_b = (p_b_given_a * p_a) + (p_b_given_not_a * p_not_a)  
  
    # 3. Calculate Posterior  
    p_a_given_b = (p_b_given_a * p_a) / p_b  
    return p_a_given_b  
  
# Medical Test Example  
print(bayes_theorem(0.01, 0.99, 0.01))    # Output: 0.5  
  
# Spam Detection Example  
print(bayes_theorem(0.4, 0.2, 0.05))    # Output: 0.727...
```

_____ Prior
_____ Likelihood
_____ False Positive Rate

Why This Matters in Data Science



Naïve Bayes Classification

The fundamental algorithm for text classification and spam filtering.



Recommendation Systems

Conditional probability engines: "Given user watched A, will they watch B?"



Risk Analysis & Causal Inference

Assessing failure probability given specific risk factors.



A/B Testing

Interpreting conversion rates: $P(\text{Conversion} \mid \text{Variant A})$.

Inter Tight: Summary & Key Formulas

1. Independence Check

Two events are independent if:

$$P(A \cap B) = P(A) \times P(B)$$

2. Conditional Probability

The probability of A given B:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

3. Bayes' Theorem

Updating beliefs with evidence:

$$Posterior = \frac{Likelihood \times Prior}{Evidence}$$

Inter Tight

Probability is not static. It evolves. As we acquire new evidence, we must update our priors to reflect a more accurate view of reality.